Engineering the quantum circuit

From Josephson quantisation to error-corrected logical memory

How did quantum behaviour become protected quantum information?

1985

Can a circuit have quantized energy levels?

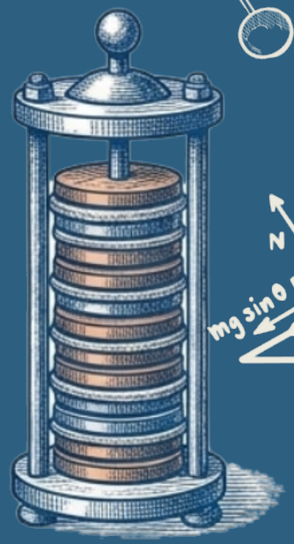
2007

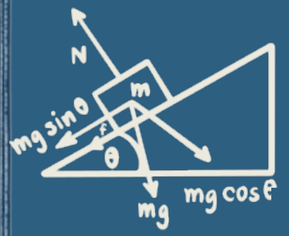
Can design reduce charge-noise sensitivity?

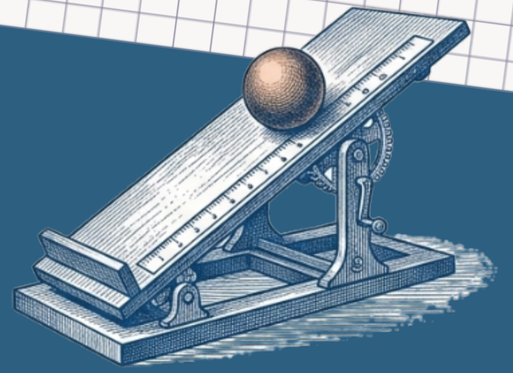
2025

Can a larger code reduce logical errors?

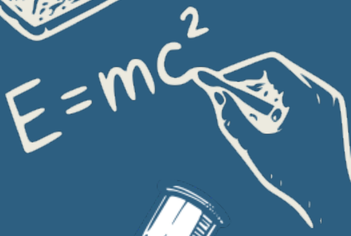


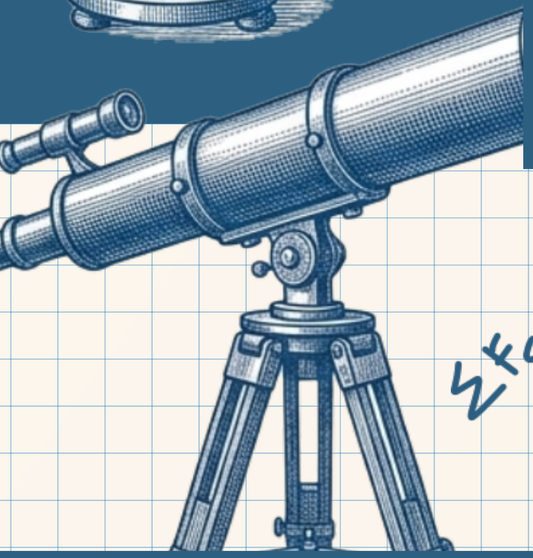


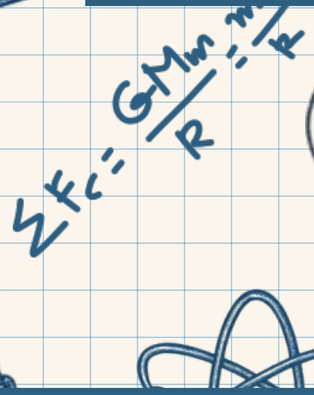








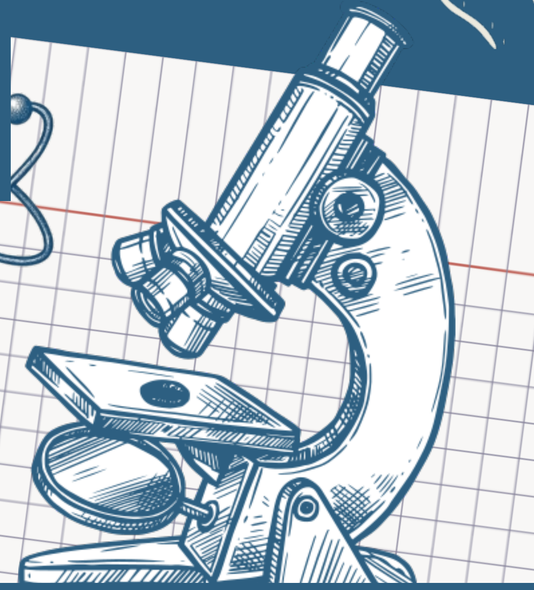












Martinis et al. (1985) · Koch et al. (2007) · Google Quantum AI and Collaborators (2025)

Contents

Introduction	03
Phase 1 · Quantized circuit levels	05
Phase 2 · Charge-stable qubits	10
Phase 3 · Logical protection	15
Conclusion	21
References	23
Backup slides	25

Introduction

Introduction

Quantum information in a circuit

Superconducting quantum computing uses engineered circuit states to carry information.

The challenge: resolve, control and preserve those states in a noisy physical device.

Why superconductivity?

Cooper pairs share a collective quantum state.

Electrical dissipation is reduced; loss still remains.

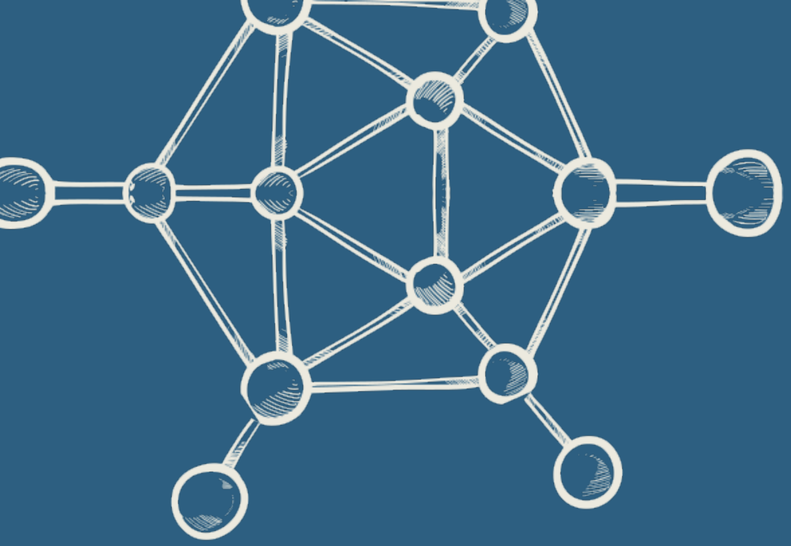
THREE GUIDING QUESTIONS IN EVERY PHASE

1 Key question

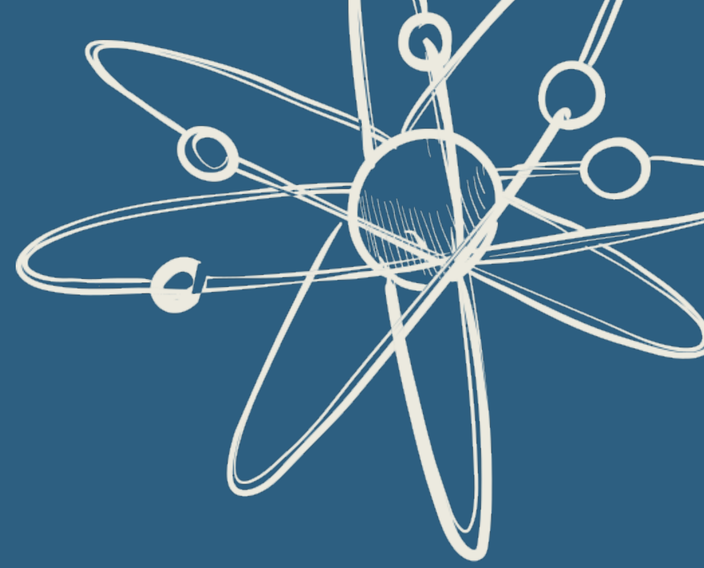
2 Methodology

3 Noise and errors

Martinis et al. (1985); Koch et al. (2007); Google Quantum AI and Collaborators (2025).





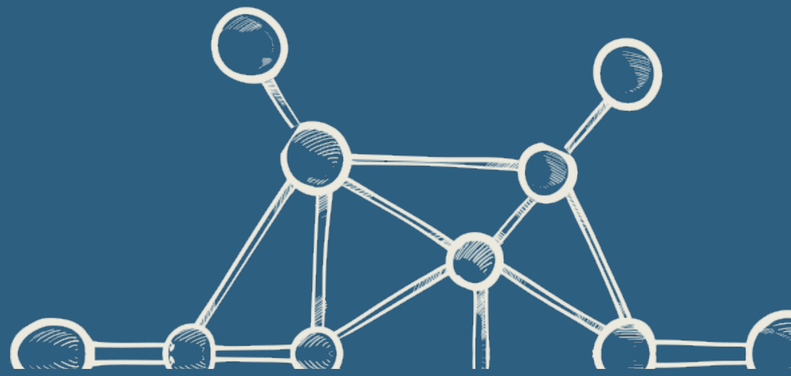


PHASE 1 · 1985

Quantized circuit levels







Unequal energy gaps enable selective control

Can a fabricated circuit exhibit discrete quantum energy levels?

LC oscillator

Capacitor + linear inductor



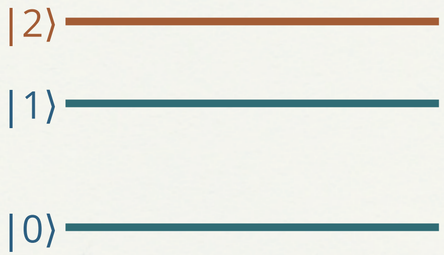
Equal gaps

The same frequency can excite both steps.

Leakage outside the chosen |0⟩, |1⟩ space

Josephson junction

Weak link → nonlinear inductance



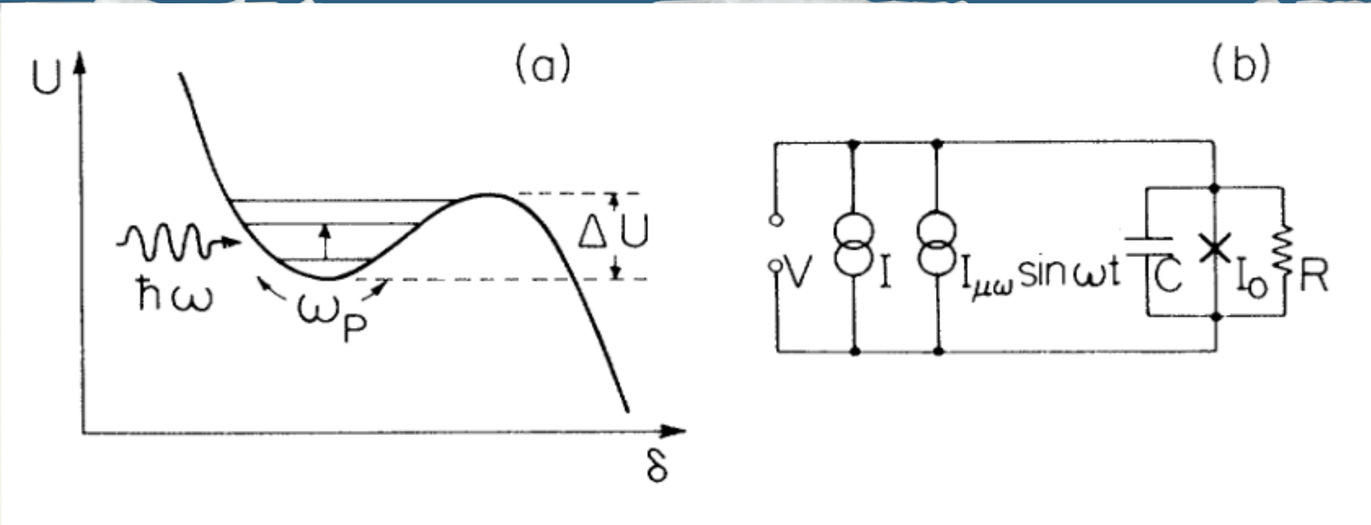
Unequal gaps

Suitable pulses can distinguish the steps.

Selective control; leakage is still possible

Schematic energy levels; |0⟩ and |1⟩ are selected circuit states.

A collective circuit variable becomes measurable



Original Fig. 1(a–b): potential and current-biased circuit

What is moving?

The collective phase difference across the junction.

$hf = E_1 - E_0$

The “particle” is not an electron moving in real space.

1 Trapped

Zero-voltage state

2 Excited

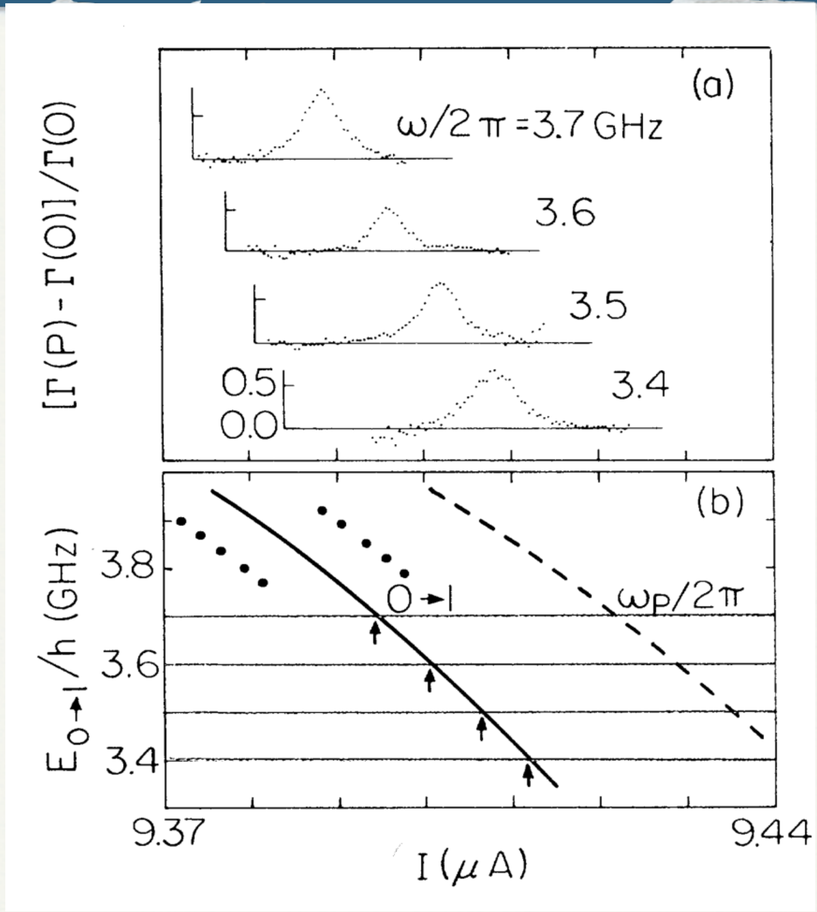
Microwaves raise the energy

3 Escaped

Measurable voltage

Martinis, Devoret & Clarke (1985), Fig. 1(a–b), p. 1543.

Microwave escape resonances support quantized levels



1 Hold frequency. Sweep current.

Panel (a): relative escape-rate enhancement.

Each trace uses one fixed microwave frequency.

2 Compare the resonance positions.

Panel (b): quantum level spacing fits the shifts.

The dashed classical prediction differs.

Agreement with quantum predictions

Independent junction parameters; 18 mK.

A peak alone is not uniquely quantum.

I: bias current (μA) · E_{01}/h: frequency (GHz)

Martinis, Devoret & Clarke (1985), Fig. 3, pp. 1544–1545.

Noise control made the measurement reliable

1 COOL

18 mK

Reduce unwanted excitation from heat.

Temperature for Fig. 3 data

2 FILTER

Quiet inputs

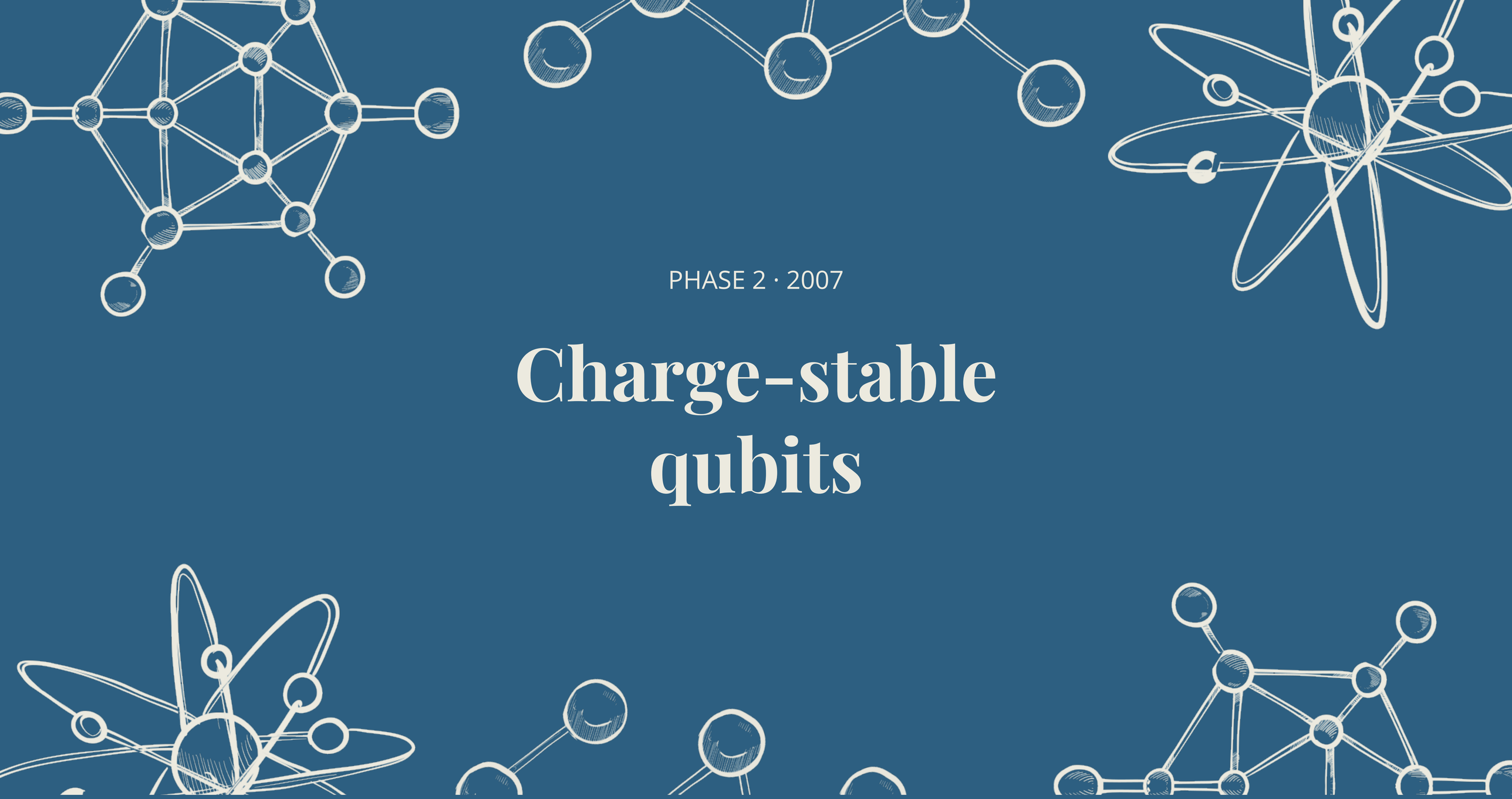
Filter electrical lines to reduce unwanted external signals.

3 RECOVER

Allow cooling

Remove the bias after switching; let heating become negligible.

These precautions revealed discrete energy levels.
Controlled manipulation and preservation of a qubit remained to be demonstrated.



PHASE 2 · 2007

Charge-stable qubits

PHASE 2 · 1 KEY QUESTION

Can a qubit be less sensitive to charge?

In a Cooper-pair box, environmental charge shifts the transition frequency.

Repeated experiments then accumulate different relative phases.

Charge fluctuates → **Frequency shifts** → **Phase is uncertain**

The environment changes.

The “clock rate” drifts.

This is dephasing.

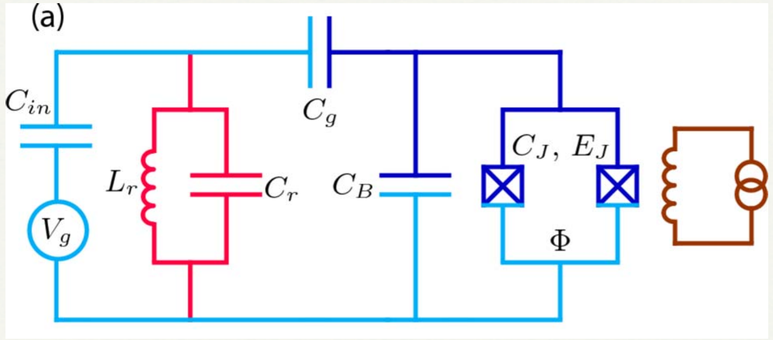
Reduce charge sensitivity while keeping unequal energy gaps.

The transmon develops from the Cooper-pair box, rather than directly modifying the 1985 device.

Koch et al. (2007), Sec. I and Sec. II A–D.

Larger capacitance reduces charge sensitivity

1 Change the circuit



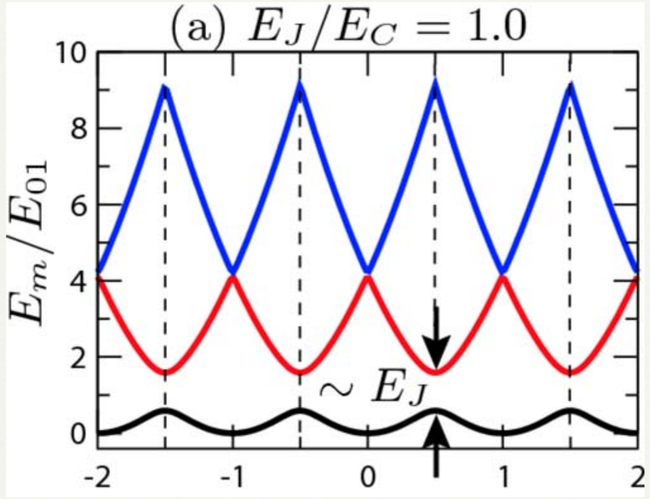
Larger CΣ → lower E_c

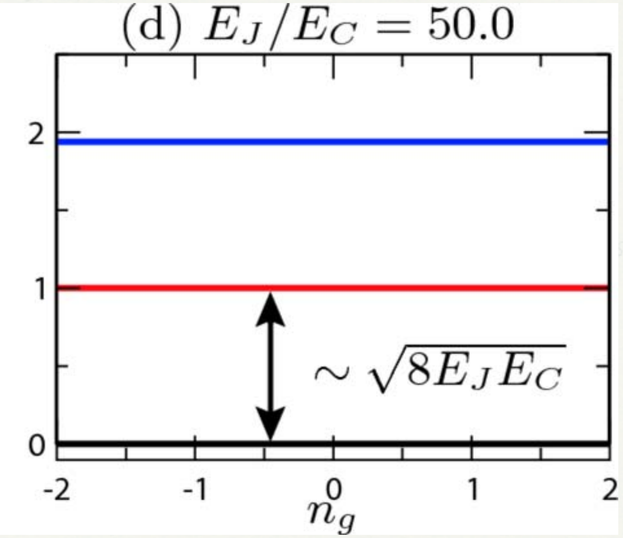
→ larger E_j/E_c

E_c: charging energy

E_j: Josephson coupling energy

2 Calculate the charge dependence





ng is offset charge; each panel uses its own E_{01} scale.

Read (a) → (d): flatter curves mean less sensitivity.

Theoretical design and numerical results: charge dispersion is strongly suppressed.

Koch et al. (2007), Fig. 1(a), Fig. 2(a,d), Sec. II A–D.

Charge stability comes with a control trade-off

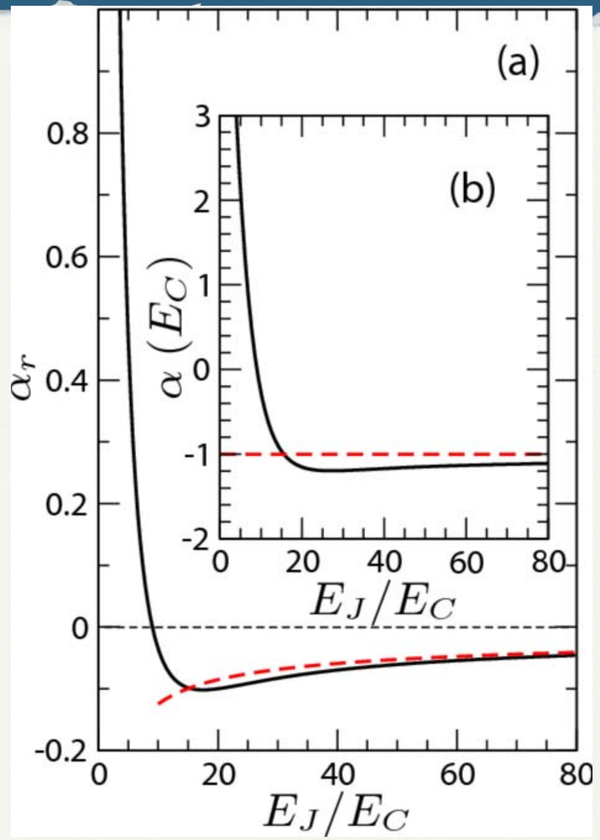
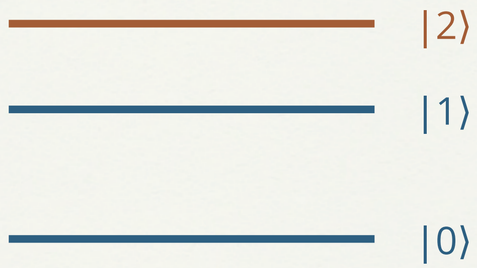


Fig. 5(a,b): theoretical calculation

Less charge sensitivity

The fluctuations remain; their effect is smaller.

Closer transition frequencies



Energy-level schematic

More leakage risk

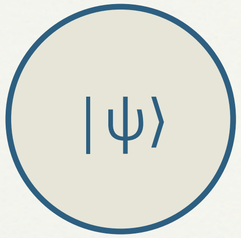
Selective driving becomes harder.

Useful anharmonicity remains.

In (a), relative spacing differences decrease; (b) approaches $-E_c$.

Better physical qubits still need logical protection

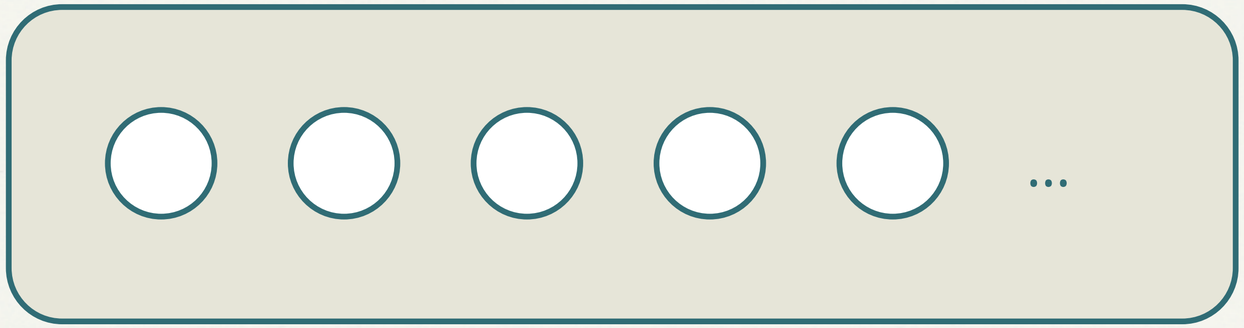
One physical qubit



Residual errors remain.

Hardware quality still matters.

One logical qubit



Encoded jointly across physical qubits.

Not copies of an unknown state. Schematic.

Relaxation · dephasing · gate and measurement errors · leakage

The 2025 experiment uses a transmon array.

Better hardware + repeated error correction

PHASE 3 · 2025

Logical protection

Can a larger code make a better memory?

A logical qubit distributes information across physical qubits.

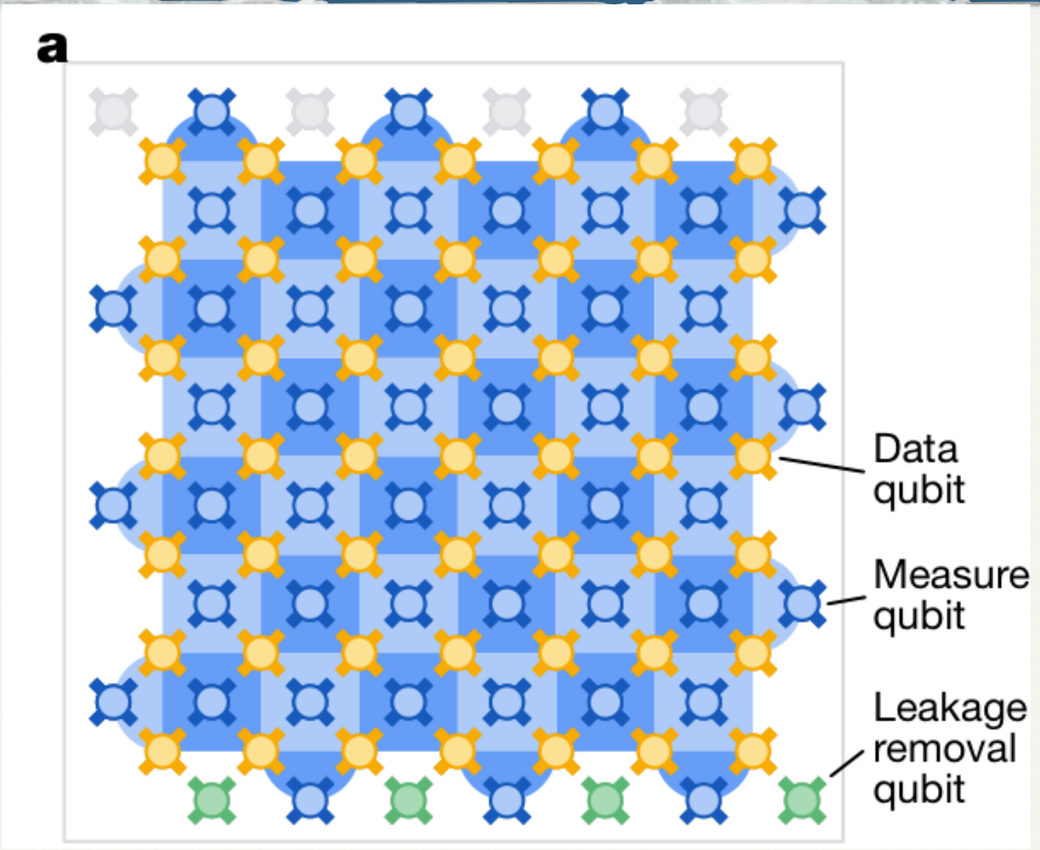
Test whether logical errors decrease as code distance grows.



Code distance d: increasing protection against combinations of physical errors

Larger codes help when physical errors are sufficiently low: below threshold.

Repeated checks reveal errors without reading the data



101 physical qubits

Distance-7 code on a 105-qubit processor

49 data + 48 measurement

+ 4 additional leakage-removal qubits

Read the pattern of checks

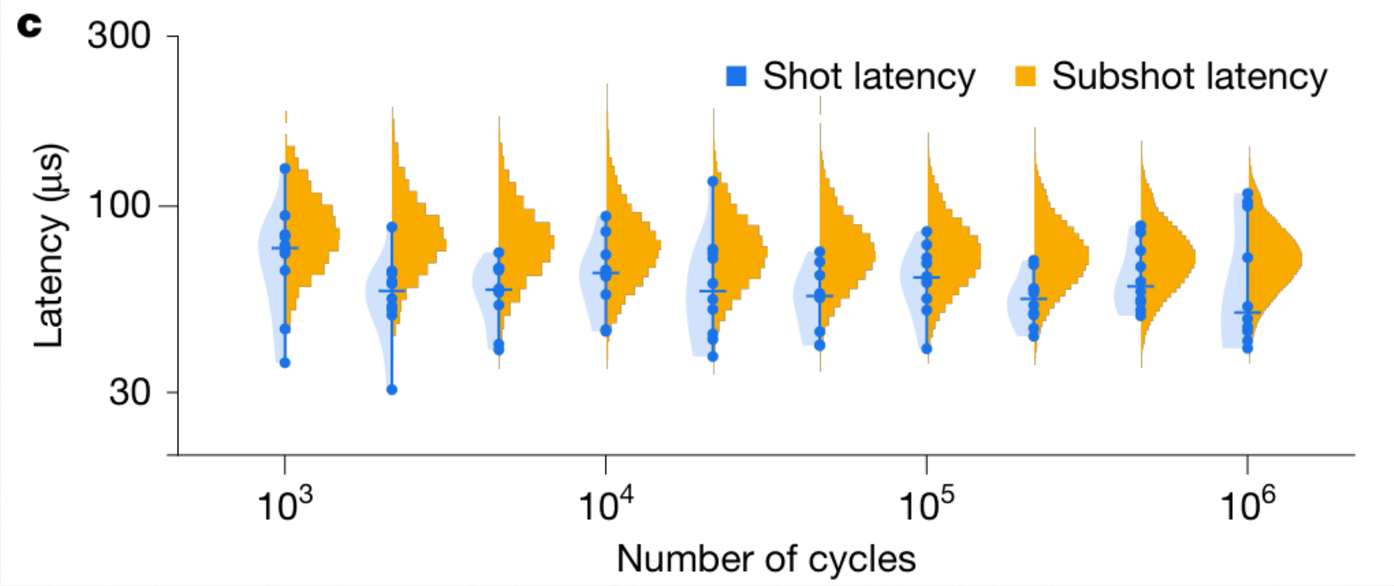
Parity checks compare collective properties.

Changes supply clues about errors.

A decoder infers the likely error pattern.

Real-time decoding must keep pace with the circuit

Separate distance-5 experiment · 72-qubit processor · sparse-blossom decoder



Read left → right: latency stays roughly constant as runs grow.

1.1 μs

Quantum correction cycle

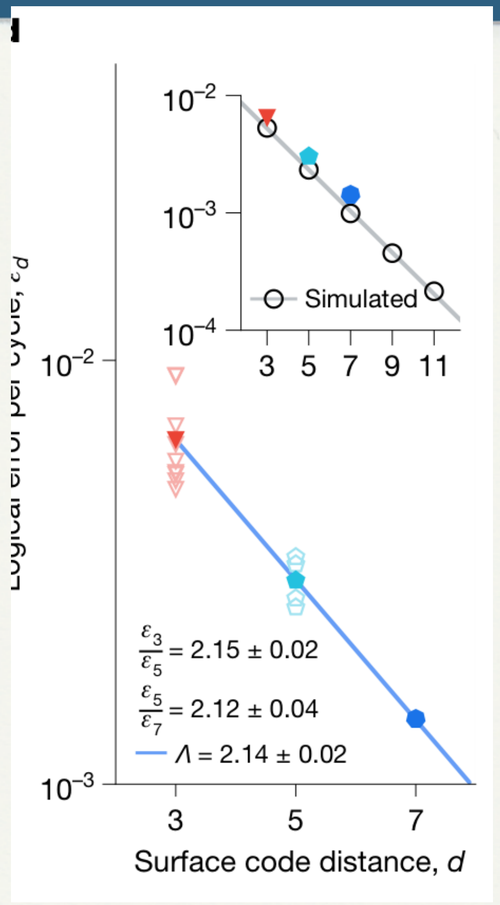
63 ± 17 μs

Mean decoding latency

Throughput keeps pace.

Latency is the output delay, not the time between cycles.

Larger codes suppress logical errors below threshold



Λ = 2.14 ± 0.02

Each +2 in code distance more than halves logical error.

Fig. 1(d), main panel: neural-network decoder; distances 3, 5, 7.

Active data-qubit leakage removal

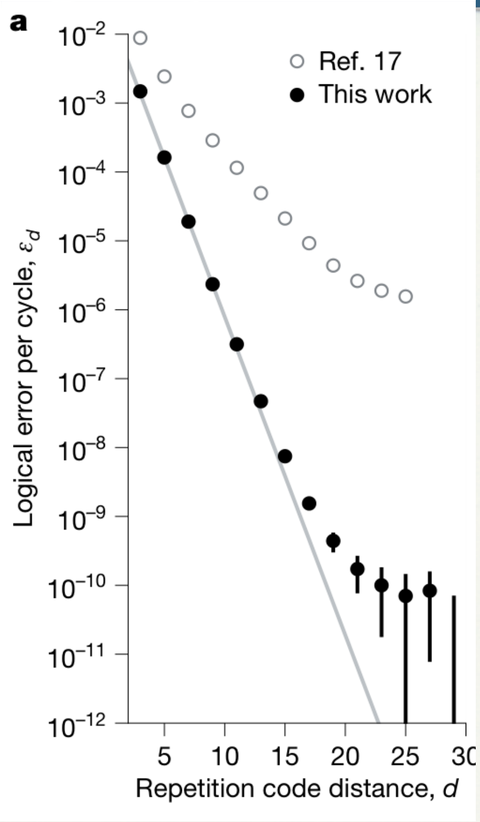
Remove leaked population to keep leakage short-lived.

291 ± 6 µs

Distance-7 logical memory lifetime

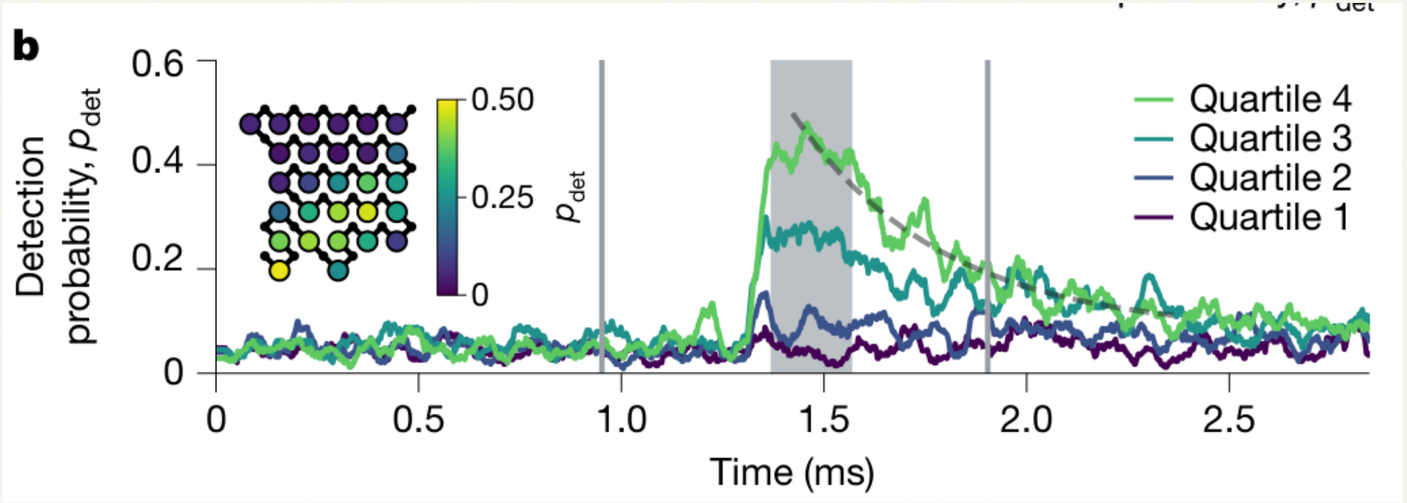
More than twice the lifetime of its best constituent physical qubit.

Rare correlated bursts expose another challenge



1 Scaling departs from the trend.

2 Many qubits are disturbed together.



Repetition codes: apparent error floor around 10^{-10} per cycle.

Roughly hourly bursts across ~30 qubits; origin unknown.

A separate repetition-code study, not a universal surface-code limit.

Conclusion

CONCLUSION

Protection now depends on the whole system

1985

Resolve the levels

Spectroscopy revealed quantization of a collective circuit variable.

Make the measurement reliable.

2007

Improve the device

The transmon reduced charge sensitivity while retaining unequal gaps.

Reduce sensitivity to disturbance.

2025

Protect the memory

Encoding, repeated checks and decoding extended logical memory lifetime.

Manage errors across the system.

Logical memory can outlive its constituent physical qubits.

Next: scale hardware and control, and understand correlated noise.

Martinis et al. (1985); Koch et al. (2007); Google Quantum AI and Collaborators (2025).

References

References

1 Observation — Martinis, Devoret & Clarke (1985)

“Energy-Level Quantization in the Zero-Voltage State of a Current-Biased Josephson Junction.”
Physical Review Letters 55, 1543–1546. DOI: 10.1103/PhysRevLett.55.1543

2 Design — Koch et al. (2007)

“Charge-insensitive qubit design derived from the Cooper pair box.”
Physical Review A 76, 042319. DOI: 10.1103/PhysRevA.76.042319

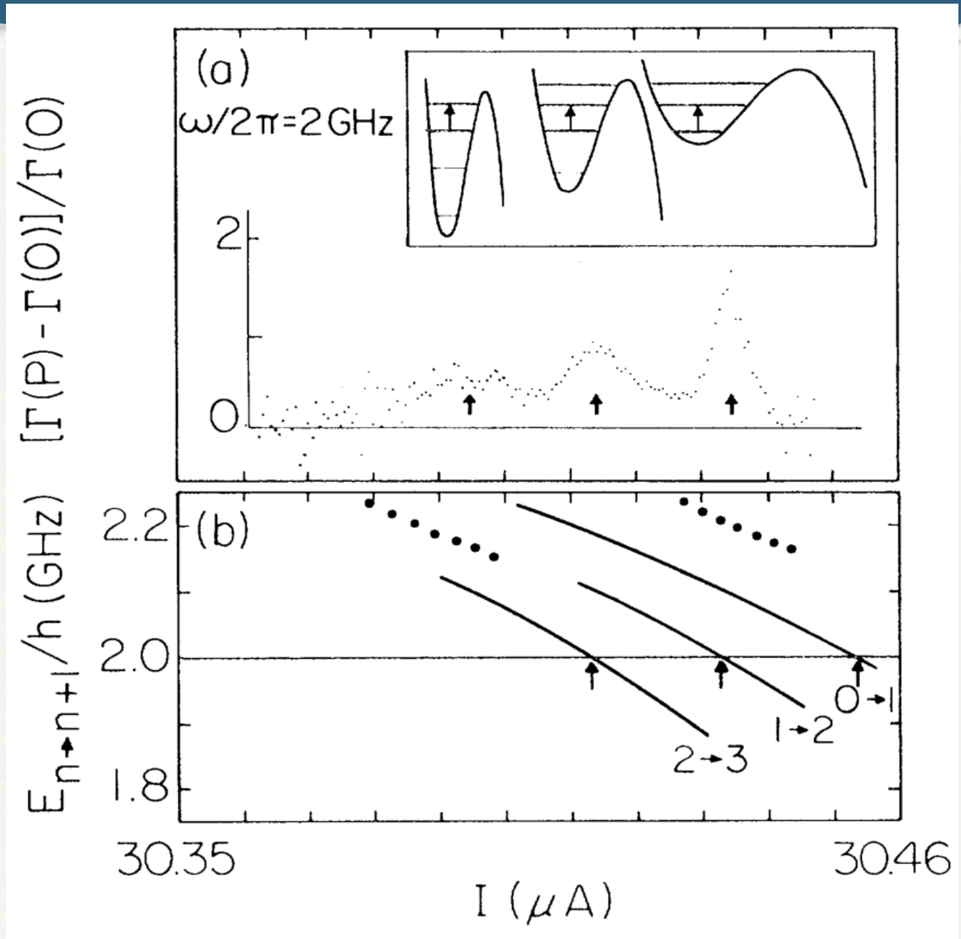
3 Protection — Google Quantum AI and Collaborators (2025)

“Quantum error correction below the surface code threshold.”
Nature 638, 920–926. DOI: 10.1038/s41586-024-08449-y

Background: Lecture 7, “Implementation”, pp. 25–27 (course slides supplied by the presenter).

Backup slides

What the 1985 evidence establishes



1 Fig. 2: several transitions

Fixed 2 GHz; sweep bias current.

Predicted 0→1, 1→2 and 2→3 resonances.

2 Fig. 3: test resonance positions

Independent junction parameters constrain theory.

Quantum and classical predictions differ.

3 Linewidth is an inference

Authors inferred coherence times up to 40 ns.

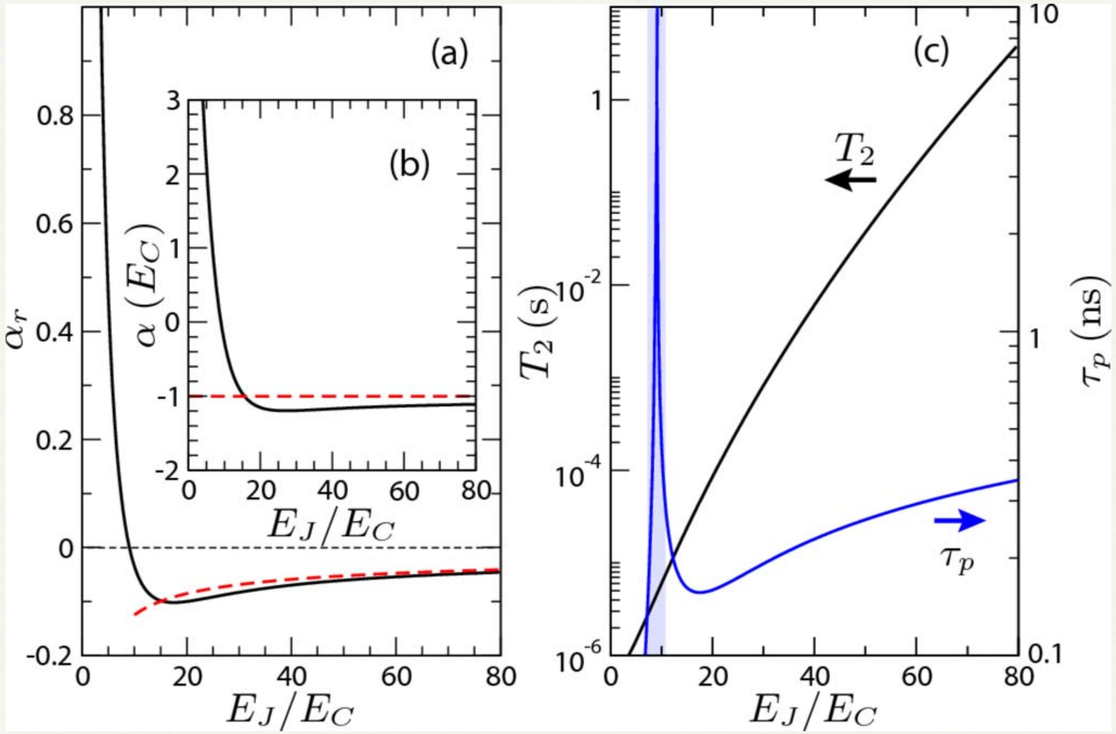
This was not a Ramsey T_2 measurement.

No controlled Rabi/Ramsey operation shown.

Martinis, Devoret & Clarke (1985), Figs. 2–4, pp. 1544–1545.

Why the transmon trade-off works

$H = 4E_c(n - ng)^2 - E_j \cos\varphi$



n: Cooper-pair number; ng: offset charge

φ: phase difference; E_j and E_c: energies

Charge dispersion

$\exp[-\sqrt{(8E_j/E_c)}]$

Leading exponential; algebraic prefactors omitted.

Anharmonicity at large E_j/E_c

$\alpha E = E_{12} - E_{01} \approx -E_c$

$\alpha r \approx -(8E_j/E_c)^{-1/2}$

αE has energy units; αE/h is in Hz.

Fig. 5: calculation, not measured coherence.

Koch et al. (2007), Eqs. (2.1), (2.5), (2.12), Fig. 5.

Keep the 2025 experiments distinct

1 Distance-7 surface-code memory

101 qubits used · neural-network decoder · Λ = 2.14 ± 0.02

Lifetime 291 ± 6 μs; best physical qubit 119 ± 13 μs; advantage 2.4 ± 0.3. [Fig. 1; pp. 921–922]

2 Distance-5 real-time decoding

72-qubit processor · sparse-blossom decoder · 63 ± 17 μs mean latency

1.1 μs correction cycle; streaming throughput keeps pace despite longer output latency. [Fig. 4; p. 924]

3 Repetition-code study up to distance 29

Apparent error floor ~10^{-10} per cycle · rare correlated bursts · unknown origin

A device- and experiment-specific limitation, not a universal thermodynamic bound. [Fig. 3; p. 923]